

SkillBarter: A Peer-to-Peer Skill Exchange Platform for Youth Employability

A Course Project Workbook

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PROJECT MANAGEMENT

Workbook

1. Project Description

SkillBarter is a peer-to-peer skill exchange platform aimed at addressing youth unemployment in India. Many young people lack job-ready skills but cannot afford formal training programmes, which are often expensive and have long waiting periods. This creates a gap between available talent and employability.

The platform uses a time-based exchange model where users teach skills they already know and earn Skill Credits. These credits can then be used to learn new skills from others. This approach enables learning without financial cost while encouraging active participation within the community.

SkillBarter also includes features such as AI-based matching to connect learners and teachers, competency assessments to validate skills, and employer integration to link users with job opportunities. In addition, it supports offline meetups and recorded learning sessions to make skill development more accessible and practical.

“The platform aims to improve employability by enabling measurable skill acquisition, verified peer learning, and direct linkage to job opportunities.”

Project Objectives

- Build an AI-powered matchmaking engine that pairs skill teachers with learners based on skill profile compatibility, geographic proximity, and mutual availability.
- Implement a transparent Skill Credit accounting system to accurately track all teaching hours, learning hours, and credit balances per registered user.
- Deploy a competency assessment engine with tamper-proof digital certificates to validate peer-learned skills for recognition by employers and partner institutions.
- Establish a teacher endorsement and reputation system enabling learners to rate instructors after every session, ensuring continuous quality control.
- Create an employer-facing job board accessible exclusively to users with verified competency certificates, directly linking skill exchange with employment outcomes.
- Develop an offline meetup coordination module for hands-on skills (bike repair, tailoring, cooking, etc.) in underserved urban and semi-urban communities.
- Build a video learning library of recorded sessions to support asynchronous learning for youth with limited or intermittent internet connectivity.
- Partner with employment exchanges, NGOs, Skill Development Centres, and SMEs to formally recognise peer-learned credentials and facilitate direct hiring pipelines.

The SkillBarter platform will expand skill accessibility to zero cost, produce verifiably employable youth, and create a self-sustaining peer learning ecosystem within communities currently excluded from formal vocational training.

PART B

2. Initiating Process Group

The Initiating Process Group formally authorises the project and establishes its strategic foundation. Per PMBOK 6th Edition, the Initiating group ensures that the project is aligned with organisational strategy, that key stakeholders are identified, and that a formal mandate exists before planning resources are committed.

This process group incorporates two core PMBOK 6th Edition processes:

- Process 4.1 – Develop Project Charter: Creates the document that formally authorises the project, defines high-level requirements, and designates the Project Manager.
- Process 13.1 – Identify Stakeholders: Identifies all individuals, groups, and organisations who may affect or be affected by the project, and documents their interests, involvement, and potential impact on project success.

3. Project Charter

Project Title	SKILLBARTER – PEER-TO-PEER SKILL EXCHANGE FOR YOUTH EMPLOYABILITY
Organization	SkillBarter Social Tech Initiative
Start Date	February 15, 2026
End Date	April 16, 2026
Project Champion	Vallala Vyshnavi
Description & Justification	India has over 75 million unemployed youth who cannot afford formal skill training (₹10,000–₹50,000 per course). Government vocational programmes have 6–12 month waiting lists. SkillBarter proposes a time-banking model where youth teach skills they already have to earn Skill Credits, which they spend learning from peers — creating a zero-cost, self-funding community skill economy that bypasses government capacity constraints and private training cost barriers.
High Level Requirements	(1) Skill Inventory Module, (2) AI Matchmaking Engine, (3) Time Tracking & Mutual Verification, (4) Skill Credit Ledger, (5) Competency Assessment Engine, (6) Digital Certificate Generator, (7) Teacher Endorsement & Reputation System, (8) Employer Job Board, (9) Learning Path Recommender, (10) Offline Meetup Coordinator, (11) Video Library
Success Criteria & Owner	1. Minimum 500 active registered youth users within 3 months (Analytics Team). 2. Minimum 300 verified skill exchanges completed by project close (QA Lead). 3. At least 80% of assessed learners earn competency certificates (Testing Lead). 4. Platform uptime of 99.9% with user registration completed within 24 hours (DevOps). 5. Minimum 20 employer partners onboarded to the job board (Business Dev Lead).
Stakeholder List	Unemployed Youth (Platform Users), Employment Exchanges, Skill Development Centres, Employers and SMEs, NGOs working on youth livelihoods, Platform Development Team
Project Budget	₹6,25,000 (inclusive of ₹50,000 Contingency Reserve and ₹25,000 Management Reserve)
Project Manager, Authority & Responsibility	Designated PM: Vallala Vyshnavi, accountable to the founding advisory board. The PM holds full authority over budget changes up to ₹20,000 per change, team assignments, and vendor selection within approved scope. Changes above ₹20,000 require CEO sign-off.
Signatures	Founder / CEO – Vallala Vyshnavi Head of Operations – Arjun Verma Date: February 15, 2026

4. Stakeholder Identification and Ranking

Stakeholder identification and ranking applies the Power-Interest Grid (Mendelow Matrix) to classify each stakeholder by their decision-making authority over the project and their personal investment in project outcomes. This classification drives the engagement strategy applied in the Stakeholder Register.

Stakeholder	Role	Power Level	Interest Level	Grid Quadrant	Engagement Strategy
Vallala Vyshnavi	Founder & CEO	High	High	Manage Closely	Engage, consult, and inform at every project stage
Arjun Verma	Project Head	High	Low	Keep Satisfied	Involve in key decisions; avoid information overload
Sneha Iyer	UX & Design Lead	Medium	High	Keep Informed	Regular updates, feedback loops, design reviews
Rahul Nair	QA & Maintenance Lead	Medium	Medium	Monitor	Periodic check-ins; no intensive engagement
NGO Partners	Community Outreach	Low	High	Keep Informed	Monthly briefings; impact reports; workshop updates
Employer Partners (SMEs)	Job Board Consumers	Low	Medium	Monitor	Quarterly partner reviews; hiring outcome reports
Govt. Employment Exchanges	Referral Partners	Low	Low	Monitor	Annual relationship updates; regulatory notices
Unemployed Youth (Users)	Platform Beneficiaries	Low	High	Keep Informed	Platform notifications; feedback surveys; community events

Power-Interest Grid Classification

The Mendelow Matrix categorises stakeholders into four quadrants based on their combination of power (authority to influence the project) and interest (degree to which the project affects them):

- Quadrant 1 – High Power, High Interest (Manage Closely): Founder/CEO. These stakeholders must be actively engaged at all stages. They have both the authority to change project direction and a strong personal stake in outcomes.
- Quadrant 2 – High Power, Low Interest (Keep Satisfied): Project Head. These stakeholders have execution authority but are primarily focused on delivery efficiency rather than mission outcomes. Keep them satisfied with milestone achievements.
- Quadrant 3 – Low Power, High Interest (Keep Informed): Design Lead, NGO Partners, Youth Users. These stakeholders are deeply affected by the project but have limited formal authority. Maintain regular communication to retain their trust and engagement.

- Quadrant 4 – Low Power, Low Interest (Monitor): Government Employment Exchanges. These stakeholders have peripheral involvement. Monitor periodically but do not invest intensive engagement resources.

5. Stakeholder Register

Stakeholder	Rank	Role	Interest / Goal	Communication Preference
Vallala Vyshnavi	High Power, High Interest	Founder & CEO	Drives strategic vision; ensures social mission alignment and long-term platform sustainability	Weekly review meetings; direct reports
Arjun Verma	High Power, Low Interest	Project Head	Manages sprint delivery, resource scheduling, and cross-team coordination	Daily standups; bi-weekly milestone reviews
Sneha Iyer	Medium Power, High Interest	UX & Design Lead	Creates accessible, mobile-first interfaces for low-literacy users; passionate about design quality	Design review sessions; prototype feedback loops
Rahul Nair	Medium Power, Medium Interest	QA & Maintenance Lead	Ensures platform stability, credit ledger security, and matchmaking algorithm accuracy	Bi-weekly QA reports; sprint retrospectives
NGO Partners	Low Power, High Interest	Community Outreach Partners	Facilitate community adoption; connect the platform with underserved youth populations	Monthly impact briefings; field visit reports
Employer Partners (SMEs)	Low Power, Medium Interest	Job Board Consumers	Access verified, certified youth talent for entry-level roles; reduce recruitment costs	Quarterly partner newsletters; job board analytics
Govt. Employment Exchanges	Low Power, Low Interest	Regulatory & Referral	Provide youth referrals; eventual MoU for formal recognition of peer certificates	Annual updates; regulatory compliance reports
Unemployed Youth (Users)	Low Power, High Interest	Platform Beneficiaries	Acquire new skills at zero cost; earn digital certificates; access job opportunities	In-app notifications; community events; surveys

“For academic purposes, the Project Manager and CEO roles are handled by the same individual.”

6. Planning Process Group

The Planning Process Group establishes the full scope, schedule, cost, quality, resource, communication, risk, and procurement baselines for the SkillBarter project. It produces the subsidiary plans that collectively form the Project Management Plan (PMP) and provides the roadmap for executing the project within approved constraints.

The Planning Process Group incorporates the following PMBOK 6th Edition processes applied to this project:

- Process 5.2 – Collect Requirements: Defines and documents stakeholder needs and functional requirements for all 11 platform modules.
- Process 5.3 – Define Scope: Establishes the Project Scope Statement, deliverables, constraints, and acceptance criteria.
- Process 5.4 – Create WBS: Decomposes project scope into the Work Breakdown Structure with Level-1, Level-2, and Level-3 elements.
- Process 6.2 – Define Activities: Identifies and documents all activities required to produce WBS deliverables.
- Process 6.4 – Estimate Activity Durations: Estimates the number of working days needed to complete each activity.
- Process 6.5 – Develop Schedule: Creates the project schedule with start/end dates, dependencies, and the critical path.
- Process 7.2 – Estimate Costs: Develops bottom-up cost estimates for all work packages.
- Process 7.3 – Determine Budget: Aggregates work-package estimates into the cost baseline and adds contingency and management reserves.
- Process 8.1 – Plan Quality Management: Defines quality standards, metrics, and control processes.
- Process 9.1 – Plan Resource Management: Identifies team roles, skills required, and the RACI Matrix.
- Process 10.1 – Plan Communications Management: Defines communication channels, frequencies, and stakeholder reporting.
- Process 11.1–11.5 – Plan Risk Management through Plan Risk Responses: Identifies, analyses, and plans responses to project risks.
- Process 12.1 – Plan Procurement Management: Defines procurement strategy, vendor selection criteria, and contract types.

7. Project Management Plan

The Project Management Plan (PMP) is the formally approved document that defines how the project will be executed, monitored, and controlled. It integrates all subsidiary plans and serves as the single source of truth for the project team and stakeholders.

Project Name	SkillBarter – Peer-to-Peer Skill Exchange for youth employability
Project Charter Reference	SkillBarter Initiative Charter v1.0 Date: February 15, 2026
Scope Statement	The project encompasses the complete design, development, testing, and pilot deployment of the SkillBarter platform across 11 modules: (1) User registration & skill inventory, (2) AI matchmaking engine, (3) Time tracking & mutual verification, (4) Skill Credit accounting, (5) Competency assessment tests, (6) Digital certificate generation, (7) Teacher endorsement & reputation system, (8) Employer job board, (9) Learning path recommender, (10) Offline meetup coordinator, (11) Video library. Pilot launch targets 500 registered youth across 3 urban areas.
In-Scope	All 11 platform modules; Android mobile application; web-based admin dashboard; cloud deployment (AWS/GCP); pilot operations in 3 cities; employer job board with 20 partner SMEs; competency certificate system.
Out of Scope	iOS application (Phase 2); rural Tier-3 city rollout (Phase 2); formal government MoUs (Phase 2); skills requiring occupational licensing (flagged, not certified); Phase 2 national scale-up.
Subsidiary Plans	(1) Scope Management Plan (2) Schedule Management Plan (3) Cost Management Plan (4) Quality Management Plan (5) Human Resource Management Plan (6) Communications Management Plan (7) Risk Management Plan (8) Procurement Management Plan (9) Stakeholder Engagement Plan
Project Approach	Agile-Waterfall Hybrid: Phases 1.1–1.2 use a waterfall approach for architecture and planning; Phases 1.3–1.7 use 2-week Agile sprints for iterative platform development and community deployment.
Key Assumptions	(1) Team availability for 4 productive hours/person/day on contracted roles. (2) Cloud infrastructure available at estimated pricing. (3) NGO partners will cooperate for community outreach. (4) Android penetration sufficient for pilot city demographics.
Key Constraints	(1) Project must complete within ₹6,25,000 total budget. (2) Hard deadline: April 16, 2026 for pilot launch. (3) Phase 1 limited to Android platform. (4) Contingency reserve fully committed after approved changes.

“Deliverables beyond the 11 core modules represent supporting outputs required for deployment and validation.”

8. Project Scope Statement

The Project Scope Statement formally documents the boundaries, deliverables, constraints, and acceptance criteria for the SkillBarter project. It is derived from the Project Charter and the Collect Requirements process, and serves as the baseline against which all scope change requests are evaluated. This section corresponds to PMBOK 6th Edition Section 5.3 (Define Scope).

Scope Statement

The SkillBarter project covers the complete design, development, testing, and pilot deployment of a peer-to-peer skill exchange platform for unemployed youth in India. The project will deliver a fully operational Android mobile application and web-based admin dashboard comprising 11 integrated modules, deployed on cloud infrastructure (AWS/GCP) and piloted with 500 registered users across 3 urban areas. The platform enables users to teach skills they possess in exchange for Skill Credits, which they spend to learn new skills from peers — creating a zero-cost, self-sustaining skill development ecosystem that links directly to employer hiring outcomes through a verified job board.

The project scope is bounded by the February 15, 2026 start date and the April 16, 2026 hard deadline. All 11 modules must reach production readiness within this window. Post-launch support and monitoring is included for 30 days beyond go-live. Activities beyond this window, including national scale-up and government MoU formalisation, are explicitly excluded from Phase 1 scope.

Deliverables

The following deliverables constitute the full set of outputs that the project must produce to be considered complete. Each deliverable maps to one or more WBS Level-3 work packages and has defined acceptance criteria:

#	Deliverable	Description	WBS Reference
1	Skill Inventory & User Profile Module	User registration, skill taxonomy (300+ skills across 12 categories), profile creation, and onboarding flow	WBS 1.3.1
2	AI Matchmaking Engine	Multi-factor algorithm pairing teachers and learners by skill, proximity, availability, and endorsement history	WBS 1.3.2
3	Time Tracking & Mutual Verification System	Session logging, dual-party confirmation, geolocation check for offline sessions, fraud detection integration	WBS 1.3.3
4	Skill Credit Accounting Ledger	Immutable credit transaction ledger with AES-256 encryption, balance management, and audit trail	WBS 1.3.4
5	Competency Assessment Engine	Timed, auto-graded competency tests mapped to NOS (National Occupational Standards) for 12 skill categories	WBS 1.4.1–1.4.2
6	Digital Certificate Generator	Tamper-proof, blockchain-anchored certificates issued upon passing competency assessments	WBS 1.4.3
7	Teacher Endorsement & Reputation System	Post-session ratings, endorsement scoring, and automatic teaching suspension for ratings below threshold	WBS 1.4.4
8	Employer Job Board & Partner Portal	Verified-alumni-only job board with 20 active SME partners;	WBS 1.5.3

		employer posting and applicant tracking interface	
9	Learning Path Recommendation Engine	Personalised skill sequence recommendations based on user profile, career goals, and market demand data	WBS 1.5.2
10	Offline Meetup Coordination Tool	Location-based meetup scheduling, venue coordination, and attendance tracking for hands-on skill sessions	WBS 1.5.1
11	Video Learning Library	Searchable library of recorded skill sessions with offline download capability for low-bandwidth users	WBS 1.3.5
12	Admin Dashboard	Web-based management interface for platform analytics, user management, credit oversight, and partner onboarding	WBS 1.3.1
13	Security-Audited Platform Build	Production build passing OWASP Top 10 audit, DPDPA 2023 compliance verification, and load testing at 500+ concurrent users	WBS 1.6.2
14	Pilot Operations Package	500 onboarded pilot users across 3 cities; 2 trained community facilitators per city; go-live sign-off documentation	WBS 1.7.1–1.7.2
15	Project Documentation Package	System architecture document, API documentation, user manuals, training guides, and lessons learned report	WBS 1.7.3

Limits and Exclusions

The following items are explicitly excluded from the Phase 1 project scope. Inclusion of any out-of-scope item requires a formal Change Request (CR) with CEO approval:

- iOS Application: Phase 1 is limited to Android. iOS development is deferred to Phase 2 pending pilot outcomes and budget availability.
- Rural and Tier-3 City Rollout: The pilot is restricted to 3 urban centres. Rural communities and Tier-3 cities are a Phase 2 expansion objective.
- Formal Government MoUs: Memoranda of Understanding with government employment exchanges and NSDC formal endorsement are Phase 2 activities.
- Occupational Licensing Scope: Skills requiring formal government licensing (electrical installation, medical practice, legal services) are flagged in the platform taxonomy but excluded from the peer certification system.
- Paid Course Integration: SkillBarter does not integrate paid third-party training content in Phase 1. All learning is peer-to-peer only.
- Multilingual Full Localisation: Phase 1 supports Hindi and Telugu for UI elements in pilot cities only. Full multilingual support across 22 Indian languages is a Phase 2 deliverable.

- Hardware Procurement for Users: The project does not provide devices to pilot users. Android penetration in pilot cities is assumed to be sufficient based on pre-pilot demographic research.
- Post-30-Day Support: Extended platform support beyond the 30-day post-launch monitoring window falls under Phase 2 operations budget.

Acceptance Criteria

The project will be formally accepted by the Founder/CEO when all of the following acceptance criteria are verified and documented by the designated owners at project close (April 16, 2026):

#	Acceptance Criterion	Verification Method	Owner
1	All 11 platform modules are live and operational in the production environment	Production deployment sign-off; smoke test passing across all modules	DevOps Engineer, QA Lead
2	Minimum 500 active registered users onboarded across 3 pilot cities	Platform analytics dashboard showing 500+ active accounts	Community Coordinator
3	Minimum 300 verified peer skill exchanges completed	Session log + mutual verification confirmation count	QA Lead
4	At least 80% of assessed learners earn a competency certificate	Certificate issuance report from assessment engine	Testing Lead
5	Platform uptime of 99.9% achieved during the first 7 days post-launch	AWS CloudWatch uptime monitoring report	DevOps Engineer
6	20 employer partners active on the job board with at least 5 posting listings	Partner onboarding log and active listing count	Business Dev Lead
7	Security audit passed with zero critical or high severity findings open at go-live	OWASP ZAP audit report; penetration test sign-off	Compliance Officer, Security Specialist
8	DPDPA 2023 compliance confirmed by the Compliance Officer	Compliance certification and data handling audit report	Compliance Officer
9	User task completion rate of at least 85% confirmed in UAT with 30 pilot users	UAT report with task completion rates per key user journey	QA Lead, Project Manager
10	Total project expenditure does not exceed the approved budget of Rs.6,25,000	Final cost report reconciled against budget baseline	Project Manager, Finance Lead
11	Complete project documentation package delivered and archived	Documentation review sign-off by Project Head	Project Head

Formal project acceptance requires written sign-off from the Founder/CEO on a Project Acceptance Certificate confirming that all 11 criteria above have been met. Criteria not met at project close will be documented as open items and transferred to the Phase 2 scope baseline.

9. Work Breakdown Structure (WBS)

The WBS decomposes the total project scope into deliverable-oriented, manageable work packages per PMBOK 6th Edition Section 5.4 (Create WBS). The 100% Rule is strictly applied: every element of project work is captured within the structure with no overlap between work packages.

WBS Levels: Level 0 = Total Project | Level 1 = Phase | Level 2 = Deliverable Group | Level 3 = Work Package

WBS ID	Phase (Level 1)	Work Packages (Level 3)
1.1	Project Initiation	1.1.1 Problem Research & User Interviews 1.1.2 Feasibility & Market Analysis 1.1.3 Stakeholder Identification & Mapping 1.1.4 Risk Assessment & Initial Mitigation Planning
1.2	Platform Design & Architecture	1.2.1 User Journey Mapping & Wireframes 1.2.2 System Architecture & Database Design 1.2.3 Matchmaking Algorithm Specification 1.2.4 Skill Credit Ledger Architecture 1.2.5 UI/UX Prototyping & Accessibility Review
1.3	Core Platform Development	1.3.1 User Registration & Skill Inventory 1.3.2 AI Matchmaking Engine 1.3.3 Time Tracking & Mutual Verification 1.3.4 Skill Credit Accounting System 1.3.5 Video Library Module
1.4	Assessment & Certification Engine	1.4.1 Competency Test Design & Build 1.4.2 Auto-Grading & Result Processing 1.4.3 Digital Certificate Generation & Verification 1.4.4 Teacher Endorsement & Reputation System
1.5	Community & Employer Modules	1.5.1 Offline Meetup Coordination Tool 1.5.2 Learning Path Recommendation Engine 1.5.3 Employer Job Board & Partner Portal 1.5.4 NGO & Employment Exchange API Integration
1.6	Testing & Validation	1.6.1 Unit, Integration & Regression Testing 1.6.2 Security & Anti-Fraud Audit 1.6.3 Community Feedback & Pilot Survey 1.6.4 Performance & Load Testing
1.7	Deployment & Project Handoff	1.7.1 Production Deployment & Go-Live 1.7.2 Community Trainer Onboarding 1.7.3 Documentation & Knowledge Transfer 1.7.4 Post-Launch Monitoring & Reporting Plan

Detailed WBS Hierarchy

The full three-level indented WBS tree is shown below:

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SkillBarter Platform [Level 0 - Total Project]
├── 1.1 Project Initiation
│   ├── 1.1.1 Problem Research & User Interviews
│   ├── 1.1.2 Feasibility & Market Analysis
│   ├── 1.1.3 Stakeholder Identification & Mapping
│   └── 1.1.4 Risk Assessment & Initial Mitigation Planning
├── 1.2 Platform Design & Architecture
│   ├── 1.2.1 User Journey Mapping & Wireframes
│   ├── 1.2.2 System Architecture & Database Design
│   ├── 1.2.3 Matchmaking Algorithm Specification
│   └── 1.2.4 Skill Credit Ledger Architecture

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- └─ 1.2.5 UI/UX Prototyping & Accessibility Review
- └─ 1.3 Core Platform Development
 - └─ 1.3.1 User Registration & Skill Inventory Module
 - └─ 1.3.2 AI Matchmaking Engine
 - └─ 1.3.3 Time Tracking & Mutual Verification Module
 - └─ 1.3.4 Skill Credit Accounting System
 - └─ 1.3.5 Video Library Module
- └─ 1.4 Assessment & Certification Engine
 - └─ 1.4.1 Competency Test Design & Integration
 - └─ 1.4.2 Auto-Grading & Result Processing
 - └─ 1.4.3 Digital Certificate Generation
 - └─ 1.4.4 Teacher Endorsement & Reputation System
- └─ 1.5 Community & Employer Modules
 - └─ 1.5.1 Offline Meetup Coordination Tool
 - └─ 1.5.2 Learning Path Recommendation Engine
 - └─ 1.5.3 Employer Job Board & Partner Portal
 - └─ 1.5.4 NGO & Employment Exchange API Integration
- └─ 1.6 Testing & Validation
 - └─ 1.6.1 Unit, Integration & Regression Testing
 - └─ 1.6.2 Security & Anti-Fraud Audit
 - └─ 1.6.3 Community Feedback & Pilot Survey
 - └─ 1.6.4 Performance & Load Testing
- └─ 1.7 Deployment & Project Handoff
 - └─ 1.7.1 Production Deployment & Go-Live
 - └─ 1.7.2 Community Trainer Onboarding
 - └─ 1.7.3 Documentation & Knowledge Transfer
 - └─ 1.7.4 Post-Launch Monitoring & Reporting Plan

10. Schedule and Milestones

Start Date: February 15, 2026 | End Date: April 16, 2026 | Critical Path: 53 working days | Total Calendar Span: 75 days

Schedule methodology: Activities 3, 4, and 5 run in parallel — the Assessment Engine (Activity 4) and Employer Portal (Activity 5) are staffed by separate sub-teams commencing during the second half of Activity 3. Activity 7 runs Monday through Wednesday (3 working days). This parallel execution reduces the critical path to 53 working days within the 75-day calendar window.

WBS Activity Schedule

#	WBS	Activity	Duration	Start	End	Milestone / Deliverable
1	1.1	Requirements & Research	10 WD	Feb 15	Feb 28	Validated problem statement, user personas, feasibility report approved
2	1.2	Platform Architecture & Design	15 WD	Mar 1	Mar 21	System architecture doc and UI/UX prototypes reviewed and approved
3	1.3	Core Backend Development	20 WD	Mar 22	Apr 14	Skill inventory, matchmaking engine, credit ledger, video library in staging
4	1.4	Assessment & Certification Engine	7 WD	Mar 23	Mar 31	Competency test system and certificate pipeline live — parallel to Activity 3
5	1.5	Employer Portal & NGO Integration	5 WD	Apr 3	Apr 7	Employer job board and NGO APIs integrated — parallel to Activity 3 final sprint
6	1.6	Testing & Security Audit	5 WD	Apr 9	Apr 13	Full regression test passed; security audit report signed off
7	1.7	Community Pilot & Deployment	3 WD	Apr 13	Apr 16	500 pilot users onboarded; trainer team briefed; go-live authorised

Project Milestone List

Milestones are listed in strict chronological order. Each milestone represents a formally reviewable deliverable gate requiring sign-off before the next phase commences.

	Milestone	Target Date	Gate Criteria
M1	Project Kickoff	February 15, 2026	Charter signed; team onboarded; sprint planning complete
M2	Architecture & Design Finalized	March 21, 2026	System architecture, DB schema, and UI prototypes approved
M3	Assessment Engine Operational	April 11, 2026	Competency tests, auto-grading, and certificates functional
M4	Core Platform MVP in Staging	April 15, 2026	All 11 modules deployed to staging environment; integration tests passing
M5	Employer Job Board Live	April 15, 2026	20 employer partners onboarded; job board accepting verified applicants
M6	Security Audit Completed	April 16, 2026	Full security audit passed; production deployment authorised
M7	Pilot Launch & Project Completion	April 16, 2026	500 users onboarded; all trainers briefed; platform live

Activity Network Diagram (Precedence Table — AON)

The Activity-on-Node (AON) precedence table below represents the project network diagram per PMBOK 6th Edition Section 6.3 (Sequence Activities). ES = Early Start (Day 0 = Feb 15), EF = Early Finish, TF = Total Float.

WBS	Activity	Predecessor	Duration	ES	EF	TF (days)	Classification
1.1	Requirements & Research	—	10	0	10	0	Critical
1.2	Platform Architecture	1.1	15	10	25	0	Critical
1.3	Core Backend Dev	1.2	20	25	45	0	Critical
1.4	Assessment Engine	1.2 (parallel to 1.3)	7	25	32	13	Non-Critical
1.5	Employer Portal	1.2 (parallel to 1.3)	5	25	30	15	Non-Critical
1.6	Testing & Audit	1.3, 1.4, 1.5	5	45	50	0	Critical
1.7	Pilot & Deployment	1.6	3	50	53	0	Critical

Critical Path: 1.1 → 1.2 → 1.3 → 1.6 → 1.7 | Total Duration: 53 working days | Activities 1.4 and 1.5 have positive float and do not affect completion if delivered before Day 45.

11. Activity Duration and Resource Estimation

Activity Effort Estimation

Effort is estimated using bottom-up estimation per WBS Level-3 work package, assuming 4 productive hours per person per working day for contracted roles.

WBS	Activity	Duration	Resources	Team Size	Hrs/Person/Day	Calc. Effort (hrs)
1.1	Requirements & Research	10 WD	Product Analyst, UX Researcher	2	4 hrs	80
1.2	Architecture & Design	15 WD	System Architect, Design Lead	2	4 hrs	120
1.3a	Matchmaking Engine Dev	10 WD	ML Engineer, Backend Dev	2	4 hrs	80
1.3b	Core Platform Dev	20 WD	Full-Stack Developers (3)	3	4 hrs	240
1.3c	Credit Ledger Dev	12 WD	Backend Dev, Security Engr.	2	4 hrs	96
1.4	Assessment & Cert Engine	7 WD	Backend Dev, Content Designer	2	4 hrs	56
1.5	Employer Portal & NGO APIs	5 WD	Backend Dev, Business Dev Lead	2	4 hrs	40
1.5b	Video Library & Async Module	8 WD	Frontend Dev, Media Specialist	2	4 hrs	64
1.6	Testing & Security Audit	5 WD	QA Tester, Cybersecurity Specialist	2	4 hrs	40
1.7	Pilot Deployment & Onboarding	3 WD	DevOps Engineer, Community Trainer	3	4 hrs	36

Gross calculated effort (sum): 852 person-hours. Net billed effort after removing parallel activity team overlap: 320 person-hours. This is the figure used in the Project Summary and the EVM baseline.

The difference between 852 calculated hours and 320 billed hours is due to parallel execution of tasks across multiple teams, where overlapping work reduces total billed effort.

Deliverables by Team

Team	Activities and Deliverables
Team 1 – Core Platform	Architecture, Matchmaking Algorithm, Core Backend, Credit Ledger, Video Library
Team 2 – Assessment & Security	Assessment Engine, Certificate System, Security Audit, Employer Portal
Team 3 – Community & Deployment	Community Training, Offline Meetup Coordination, User Onboarding, Helpdesk

Resource Estimation Table

The following table lists all human resources required for the project, their quantity, engagement duration, required skills, and assigned activities.

Resource	Qty	Duration	Key Skills Required	Assignment
Product Analyst	1	1 month	Analytical skills, PM tools	Full-time – Activity 1.1
UX Researcher	1	1 month	User research, personas	Full-time – Activity 1.1
System Architect	1	1.5 months	System design, API architecture	Full-time – Activity 1.2
ML Engineer	1	2 months	TensorFlow Lite, NLP, recommendation	Full-time – Activities 1.3a, 1.4
Full-Stack Developer	3	3 months	React Native, Node.js, PostgreSQL	Full-time – Activity 1.3b
Backend Dev / Security	2	2.5 months	Node.js, Redis, OWASP, AES-256	Full-time – Activities 1.3c, 1.6
Content Designer	1	1 month	Instructional design, assessment build	Part-time – Activity 1.4
Frontend Developer	1	1 month	React Native, PWA, accessibility	Full-time – Activity 1.5b
Media Specialist	1	1 month	Video editing, compression	Part-time – Activity 1.5b
UI/UX Designer	1	1.5 months	Figma, WCAG 2.1, mobile-first design	Full-time – Activities 1.2, 1.3b
QA Tester	1	2 months	Jest, Postman, test case design	Full-time – Activities 1.6, 1.7
Cybersecurity Specialist	1	1.5 months	OWASP ZAP, penetration testing	Contract – Activities 1.3c, 1.6
DevOps Engineer	1	1 month	AWS/GCP, CI/CD, k6 load testing	Full-time – Activity 1.7

Community Trainer	2	1 month	Facilitation, community engagement	Contract – Activity 1.7
Employer Relations	1	2 months	Employe relation partnership	Part-time – Activity 1.5

“The reduction from 852 to 320 person-hours is due to parallel task execution across multiple teams, reducing billed effort while maintaining total workload.”

12. Risk Management Plan

The Risk Management Plan defines the processes, tools, and responsibilities for identifying, analysing, responding to, and monitoring risks throughout the SkillBarter project lifecycle. It is developed per PMBOK 6th Edition Chapter 11 (Project Risk Management) and applies to all project phases from initiation through to pilot deployment.

Risk Methodology	Qualitative Risk Analysis (probability × impact scoring) with Risk Register tracking. Quantitative analysis applied to the top 3 risks using Expected Monetary Value (EMV) calculations.
Risk Scoring Scale	Probability: High (>60%), Medium (20–60%), Low (<20%). Impact: High (major schedule/cost/quality effect), Medium (moderate effect), Low (minor, manageable effect). Overall Risk Score = Probability × Impact.
Risk Owner Assignment	Each risk is assigned a named owner responsible for monitoring, executing the response strategy, and escalating when a trigger event occurs.
Risk Review Frequency	Risk Register reviewed bi-weekly at milestone review meetings. New risks can be logged at any time. Trigger events must be escalated to the PM within 24 hours of detection.
Risk Thresholds	Risks rated High × High are escalated to the CEO immediately. Risks rated Medium × High or High × Medium require PM-level action within 48 hours. Low risks are monitored by the assigned owner.
Contingency Reserve	₹50,000 (~8.7% of base budget ₹5,75,000) is reserved for identified risks. Each drawdown must be documented against a specific risk ID and approved by the PM.
Management Reserve	₹25,000 held for unknown-unknown events (force majeure, regulatory changes). Accessible only by the CEO with a formal Change Request.
Tools	Risk Register (maintained in project management tool), Risk Heatmap, PMBOK 11.4 Probability-Impact Matrix, bi-weekly risk dashboard reports.

13. Risk Matrix

All project risks are identified and classified using the PMBOK 11.4 Probability-Impact Matrix. Each risk is rated on a scale of High, Medium, or Low for both Probability (likelihood of occurrence) and Impact (consequence severity). The matrix below provides both the visual grid and the full risk list.

Probability-Impact Matrix

Cell colour indicates overall risk severity: High Probability × High Impact = VERY HIGH (immediate CEO escalation required).

Probability \ Impact	Low Impact	Medium Impact	High Impact
High Probability	MEDIUM — R-05	HIGH — R-04	VERY HIGH — R-01, R-02
Medium Probability	LOW — R-09	MEDIUM — R-03, R-07	HIGH — R-06
Low Probability	LOW — R-08, R-10	LOW — R-08, R-10	MEDIUM — R-06

Risk Classification Table

Risk ID	Risk Description	Type	Probability	Impact	Planned Response	Residual Rating
R-01	Poor Teaching Quality	Threat	High	High	Mandatory endorsements; rating threshold policy	Medium
R-02	Skill Credit Inflation	Threat	High	High	Algorithm credit caps; teacher-learner ratio control	Medium
R-03	Fake Session Logging	Threat	Medium	High	Mutual verification; geolocation; AI fraud detection	Medium
R-04	Employer Non-Recognition	Threat	Medium	Medium	NSDC alignment; NOS-mapped assessments; employer advisory board	Low
R-05	User Dropout Post-Credits	Threat	High	Medium	Gamification; milestone badges; re-engagement campaigns	Low
R-06	Data Privacy Breach	Threat	Low	High	AES-256 encryption; DPDPA	Low

					compliance; bi-annual audits	
R-07	Low Community Adoption	Threat	Medium	High	NGO ambassador network; employment exchange integrations	Medium
R-08	Platform Downtime	Threat	Low	Medium	Cloud redundancy; CI/CD pipeline; uptime monitoring	Low
R-09	Matchmaking Accuracy Drop	Threat	Medium	Medium	Weekly model benchmarking; retraining pipeline	Low
R-10	API Integration Failure	Threat	Low	Medium	Retry logic; fallback connectors; staging validation	Low

14. Risk Register

The Risk Register is the living document that tracks all identified risks, their current ratings, response strategies, owners, and status throughout the project lifecycle. Per PMBOK 11.6, risks with post-response ratings of Medium or High require formal Contingency Plans.

ID	Risk	Pre-Response Rating	Response Strategy	Owner	Post-Response Rating	Current Status
R-01	Poor Teaching Quality	H	Mitigate: Mandatory learner endorsements after every session; automatic suspension when average rating falls below 4.0/5.0; peer mentor assignment for remediation	QA Lead, Community Coordinator	M	Rating Monitoring Active
R-02	Skill Credit Inflation	H	Prevent: Algorithm enforces credit supply caps per skill category; balanced teacher-learner ratio dashboard with automatic throttling when ratio exceeds 2:1	ML Engineer, Backend Dev	M	Mitigation In Progress
R-03	Fake Session Logging	H	Prevent: Mutual time verification (both parties confirm); GPS-based geolocation confirmation for offline sessions; AI anomaly detection on credit-earning velocity	DevOps, Security Engineer	M	Fraud Detection Live
R-04	Employer Non-Recognition	M	Enhance: Build employer advisory board of 20 SME partners; map competency tests to NOS (National Occupational Standards); pursue NSDC co-branding endorsement	Business Dev Lead, PM	L	Partnership Outreach Active
R-05	User Dropout Post-Credits	M	Mitigate: Gamification features (badges, leaderboards); milestone-based certificate incentives; push notification re-engagement campaigns	Community Coordinator	L	Monitoring Weekly
R-06	Data Privacy Breach	M	Prevent: AES-256 encryption at rest and in transit; full DPDPA 2023 compliance; bi-annual independent penetration testing; no third-party data sale policy	Compliance Officer	L	Audit Cleared

R-07	Low Community Adoption	H	Mitigate: Pre-launch NGO ambassador network in 3 pilot cities; integration with 5 district employment exchanges for referral traffic; social media onboarding campaign	Outreach Team, PM	M	Onboarding In Progress
R-08	Platform Downtime	L	Accept with mitigation: AWS multi-AZ deployment; automated health checks; on-call DevOps rotation; 99.9% uptime SLA with vendor penalty clause	DevOps Engineer	L	SLA Monitoring Active
R-09	Matchmaking Accuracy Drop	M	Mitigate: Weekly accuracy benchmarking against 85% target; automated model retraining pipeline triggers when accuracy drops below threshold	ML Engineer	L	Monitoring Underway
R-10	API Integration Failure	L	Accept with mitigation: Retry logic and fallback connectors deployed on all third-party API connections; staging integration testing before production onboarding	API Engineer, QA Lead	L	Integration Stable

Risk Contingency Plan

The following contingency plans are activated when the defined risk trigger event is observed. These are pre-approved responses requiring no additional sign-off to execute.

Risk ID	Risk Trigger Event	Contingency Steps
R-01	Teacher avg. rating < 3.5 for 3+ consecutive sessions	Suspend account; trigger re-assessment; assign senior peer mentor before reinstatement
R-02	Credit transaction velocity exceeds defined cap for any skill category	Activate credit freeze protocol; reduce per-session earning rate; open new demand categories
R-03	Fraud patterns detected by monitoring algorithm	Freeze implicated credits; trigger manual security review; enforce mandatory video verification
R-04	< 10 employer partners by project Day 60	Activate NGO direct hiring pipeline; issue blockchain-anchored certificates for credibility
R-05	Weekly active ratio drops below 60% of registered users	Deploy bonus-credit re-engagement campaign; introduce time-limited skill challenges
R-06	Security breach reported or detected	Incident response: isolate data, notify users within 72 hours, engage forensics firm, regulatory notification
R-07	< 100 active users by end of Project timeline	Emergency ambassador deployment; partner with 5 additional Employment exchanges

15. Quality Management Plan

The Quality Management Plan defines the quality standards, responsibilities, assurance processes, and control mechanisms that will ensure the SkillBarter platform delivers accurate matchmaking, reliable credit transactions, and verifiable competency outcomes. It is developed per PMBOK 6th Edition Chapter 8 (Project Quality Management).

Quality Philosophy	Quality is built into the platform through design-phase standards, sprint-based testing, and continuous community feedback — not inspected in at the end.
Quality Standards	ISO/IEC 25010 (Software Quality), WCAG 2.1 Level AA (Accessibility), OWASP Top 10 (Security), NOS mapping for assessment frameworks (competency validity).
Quality Assurance Approach	Continuous peer review of algorithm outputs; weekly credit ledger reconciliation; bi-weekly platform performance benchmarking against KPIs; feedback collected after every completed skill exchange.
Quality Control Approach	Automated regression testing after every code deployment; manual session quality audits (random 10% sampling); teacher suspension protocols triggered automatically when endorsement scores fall below threshold.
Quality Improvement Approach	Monthly retrospectives identify improvement opportunities; algorithm weighting reviewed quarterly; employer advisory board feedback drives assessment redesign.
Technology Stack	React Native (Mobile), Node.js + Express.js (Backend), PostgreSQL (DB), Redis (Credit Ledger Cache), AWS S3 + CloudFront (Video), TensorFlow Lite (ML Matching). Testing: Jest, Postman, OWASP ZAP, k6.

Quality Roles and Responsibilities

Role	Quality Responsibility
Project Manager	Owns the Quality Management Plan; ensures all development milestones meet defined quality benchmarks; approves quality gate sign-offs.
QA Lead	Designs and executes all test cases for matchmaking accuracy, credit ledger integrity, and system performance; manages the QA team.
Platform Engineer (DevOps)	Monitors backend uptime, API response time, and data accuracy across all platform modules in real time.
Community Quality Coordinator	Collects and analyses learner feedback, endorsement scores, and teaching effectiveness data from all pilot sessions.
Compliance Officer	Ensures full platform adherence to India's Digital Personal Data Protection Act (DPDPA) 2023 and relevant educational content regulations.
ML Engineer	Monitors and maintains matchmaking algorithm accuracy; owns the model retraining pipeline and accuracy reporting.

16. Quality Metrics and Quality Control

Quality Standards and Responsible Parties

Quality Goal	Target	Responsible	Control Mechanism
Matchmaking Accuracy	85%+ match satisfaction rate	Platform ML Team	Weekly algorithm output review
Credit Ledger Integrity	100% transaction accuracy; < 0.01% error rate	Backend Dev, QA Lead	Daily automated + weekly manual reconciliation
System Uptime	99.9% monthly SLA	DevOps Engineer	Real-time monitoring dashboard
Certificate Validity	90%+ assessment reliability score; NOS alignment	QA Lead, Content Lead	Monthly certificate & assessment audit
User Satisfaction	85%+ positive feedback rating	Community Coordinator	Post-session survey after every 10 exchanges
Teacher Endorsement Rate	70%+ of active teachers with avg. score > 4.0	QA Lead	Bi-weekly endorsement score review
Issue Resolution Time	< 48 hours from fault report to resolution	PM, DevOps, QA	Real-time ticketing system; SLA monitoring

Key Performance Indicators (KPI Table)

KPI	Definition	Target	Measurement Frequency
Matchmaking Score	% of sessions rated 4/5 or above by both parties	85%+	Weekly
Credit Ledger Accuracy	Error rate in Skill Credit transaction processing	< 0.01%	Daily
Platform Uptime	System availability during active hours	99.9%	Real-time
Certificate Issue Rate	% of assessed learners earning a certificate	80%+	Monthly
Teacher Endorsement Rate	% of teachers with avg endorsement > 4.0	70%+	Bi-weekly
Issue Resolution Time	Time from fault report to confirmed resolution	< 48 hours	Per incident
User Retention Rate	% of registered users active after 30 days	60%+	Monthly
Session Completion Rate	% of matched sessions actually completed	75%+	Weekly

Quality Control Actions

Action	Responsible Person(s)	Frequency	Measurement Criteria
Matchmaking Quality Review	QA Lead, ML Engineer	Weekly	Algorithm accuracy vs. 85% target; session completion rate
Credit Ledger Reconciliation	Backend Dev, Security Engineer	Daily (auto) + Weekly (manual)	Transaction error rate vs. < 0.01% target
Platform Performance Test	DevOps Engineer, QA Team	After every deployment	API latency < 200ms; uptime 99.9%; error rate < 0.1%
Community Feedback Analysis	Community Coordinator, Support Team	After every 10 sessions	Average session rating; endorsement completion %
Certificate & Assessment Audit	Content Lead, Compliance Officer	Monthly	Assessor validation score; NOS employer alignment %
Security & Compliance Audit	Compliance Officer, Security Specialist	Bi-annual	OWASP Top 10 pass/fail; DPDPA compliance status

Quality Reporting Plan

The Quality Reporting Plan specifies how quality data is captured, analysed, and communicated to stakeholders per PMBOK 6th Edition Section 8.3 (Control Quality).

Report Type	Content	Owner → Recipient	Frequency
Weekly QA Dashboard	Bug counts by severity tier, uptime %, API latency, matchmaking accuracy	QA Lead → Project Manager	Every Monday morning
Bi-Weekly Session Quality Report	Learner satisfaction scores, endorsement rates, teacher performance flags	Community Coordinator → Project Head	Every 2nd Friday
Monthly Quality Audit Report	Full KPI vs. target review; corrective action log; risk trigger status	QA Lead + Compliance Officer → CEO	Last working day of each month
End-of-Phase Quality Gate Report	Formal sign-off confirming all quality criteria met before phase closure	Project Manager → All Stakeholders	At each project milestone gate

17. Design Phase

The Design Phase (WBS 1.1–1.2) encompasses all planning, architecture, and prototyping activities required before development begins. This phase produces the foundational design documents, system blueprints, and validated UI/UX prototypes that guide the entire development lifecycle. A quality gate review is conducted before the phase closes to ensure design sign-off.

Phase Duration: February 15 – March 21, 2026 (25 working days) | Phase Objective: Produce approved architecture, algorithm specifications, database schema, UI/UX prototypes, and all design documents required to commence development.

Design Area	Quality Activities	Responsible	Quality Metrics
User Profiling & Skill Inventory	Define user profile data model, skill taxonomy (300+ skills in 12 categories), and onboarding flow design	Design Lead	Profile completion rate %; taxonomy validation score by domain expert
Matchmaking Algorithm Design	Specify multi-factor matching criteria: skill level, proximity, availability, endorsement history; define bias safeguards	ML Engineer	Algorithm accuracy benchmark %; bias audit score; F1 precision-recall metric
Skill Credit Ledger Architecture	Design immutable transaction log with AES-256 encryption, anti-fraud rules, balance reconciliation logic, and audit trail	Backend Dev	Ledger integrity score; fraud test pass/fail; reconciliation accuracy 100%
Competency Test Framework Design	Create assessment frameworks mapped to NOS (National Occupational Standards) industry skill standards	Content Lead	Assessor validation score; employer alignment %; NOS coverage %
System Architecture & DB Design	Define microservices-based backend architecture, PostgreSQL schema, Redis caching strategy, API contract specifications	System Architect	Architecture review sign-off; DB query performance benchmark < 100ms
UI/UX Prototyping & Accessibility	Build mobile-first, WCAG 2.1-compliant Figma prototypes for all 11 modules; conduct low-literacy usability testing	Design Lead	Usability score %; WCAG 2.1 AA compliance; task completion rate > 85%

Design Phase Gate Criteria

- System architecture document reviewed and approved by the Project Head.
- Database schema reviewed and validated by the Backend Development Lead.
- Matchmaking algorithm specification reviewed by the ML Engineer with documented accuracy targets.
- All UI/UX prototypes tested with a minimum of 10 target users; task completion rate must exceed 85%.
- Skill taxonomy validated by at least one domain expert per skill category before development commences.

18. Planning Phase

The Planning Phase runs concurrently with and immediately after the Design Phase. It establishes all quality control frameworks, development infrastructure, sprint backlogs, and team training required to execute the project efficiently. A planning gate review ensures that all pre-development conditions are met before the first development sprint commences.

Phase Duration: March 3 – March 21, 2026 (concurrent with Design Phase) | Phase Objective: Establish QA processes, sprint planning, infrastructure provisioning, compliance architecture, and full team readiness for development kickoff.

Planning Area	Quality Activities	Responsible	Quality Metrics
QA Strategy & Control Plan	Define QA process covering matchmaking accuracy, credit ledger integrity, and all 11 module user interactions	QA Lead	QA plan formally approved; test coverage target $\geq 90\%$
Risk Identification (Quality Focus)	Map algorithm bias risks, fraud pattern vulnerabilities, dropout trigger scenarios, and data integrity edge cases	Risk Analyst	Risk registry completeness score; all High risks have mitigation owners
Compliance & Policy Audit	Verify DPDPA 2023 compliance architecture, youth data protection safeguards, and content regulatory requirements	Compliance Officer	Audit pass score; DPDPA compliance certification obtained
Sprint Planning & Backlog Creation	Create product backlog for all 11 modules; assign story points; define Definition of Done for each sprint	Project Head	Sprint velocity baseline established; backlog groomed and prioritised
Infrastructure Setup	Provision AWS/GCP environments: development, staging, and production; configure CI/CD pipeline (GitHub Actions)	DevOps Engineer	All three environments live; CI/CD pipeline passing automated builds
Team Training & Onboarding	Train development, QA, and community teams on platform tools, coding standards, DPDPA protocols, and facilitation methods	Training Coordinator	Training completion rate 100%; post-assessment score $\geq 80\%$

Planning Phase Gate Criteria

- QA strategy and control plan formally approved by the Project Manager.
- All three environments (dev, staging, production) provisioned and CI/CD pipeline passing automated builds.
- Product backlog groomed and sprint 1 planning complete with story points assigned.
- All team members have completed onboarding training with post-assessment scores $\geq 80\%$.
- DPDPA compliance architecture validated by the Compliance Officer before any user data collection commences.

19. Execution Phase

The Execution Phase (WBS 1.3–1.7) covers all platform development, testing, pilot deployment, and community onboarding activities. It runs from March 24 through April 16, 2026. The execution phase uses 2-week Agile sprints with continuous integration, sprint reviews, and retrospectives to maintain quality and alignment with project goals.

Phase Duration: March 24 – April 16, 2026 (25 working days on critical path; 53 WD including design/planning phases) | Phase Objective: Build, test, deploy, and operate all 11 platform modules; onboard 500 pilot users and 20 employer partners.

Execution Area	Quality Activities	Responsible	Quality Metrics
System Testing & Validation	Unit, integration, and end-to-end testing for all 11 platform modules; regression suite runs after every deployment	QA Team & Dev Team	Bug severity count by tier; test pass rate $\geq$ 95%; regression suite < 10 failures
Real-Time Performance Monitoring	Monitor live matchmaking outputs, credit transactions, video library load, and API latency during pilot operations	DevOps Engineer	Uptime $\geq$ 99.9%; API response time < 200ms; daily error rate < 0.1%
Community Feedback Collection	Collect post-session ratings, learner satisfaction scores, and teacher endorsements from all pilot participants	Community Coordinator	Average session rating $\geq$ 4.0/5.0; endorsement completion rate $\geq$ 80%
Security & Compliance Checks	Penetration testing across all endpoints; OWASP Top 10 audit; data encryption verification; fraud pattern review	Compliance & Security Officers	OWASP Top 10 audit pass; zero critical vulnerability open at go-live; audit pass/fail ratio
Continuous Improvement	Use sprint retrospectives and community feedback to iteratively improve platform features and UX each sprint	QA Lead	Number of improvements deployed per sprint; issue recurrence rate < 5%
Employer Partner Onboarding	Activate employer job board; onboard 20 SME partners; configure verified access controls for certified alumni	Business Dev Lead	20 partners onboarded by April 16; job board listings active; access controls verified

Execution Phase Gate Criteria

- All 11 platform modules pass unit and integration testing with a test pass rate $\geq$ 95%.
- Security audit completed with zero critical or high vulnerabilities open at production deployment date.
- 500 pilot users onboarded with at least 300 completing one verified skill exchange.
- 20 employer partners active on the job board with at least 5 posting active job listings.
- Platform uptime maintained at $\geq$ 99.9% during the first 7 days post-launch.

20. Human Resource Management Plan

Project Title: SkillBarter – Peer-to-Peer Skill Exchange for Unemployed Youth | **Version:** 1.0 | **Date:** February 15, 2026

The Human Resource Management Plan defines how project roles are structured, how team members are acquired, trained, managed, and released, and how performance is evaluated. It is developed per PMBOK 6th Edition Section 9 (Project Resource Management).

Core Team Roles and Responsibilities

- **CEO / Founder – Vyshnavi Vallala:** Provides strategic vision and overall project direction; secures stakeholder and employer partnerships; ensures the platform's social mission is embedded in every product decision; approves budget changes above ₹20,000.
- **Head of Project – Arjun Verma:** Manages sprint planning and resource scheduling; coordinates delivery across development, QA, and community teams; conducts bi-weekly performance reviews; escalates risks and issues to the CEO.
- **Design Lead – Sneha Iyer:** Leads UX research, wireframing, Figma prototyping, and mobile-first interface design; ensures WCAG 2.1 accessibility standards for low-literacy and first-time smartphone users are met across all modules.
- **Testing & Maintenance Lead – Rahul Nair:** Manages all QA cycles; oversees matchmaking accuracy testing; ensures the credit ledger, certificate system, and all API integrations are fully functional, secure, and performant at all times.

Staff Acquisition Strategy

- Recruit full-stack developers with proven experience in React Native mobile development and real-time transaction systems.
- Engage an ML engineer on a freelance milestone contract to build and fine-tune the peer matchmaking algorithm and learning path recommendation engine.
- Hire a UI/UX designer with an edtech or social impact portfolio for accessible, mobile-first interface development (Figma-to-React Native handoff required).
- Partner with established community development NGOs to source trained facilitators for offline meetup coordination and pilot user onboarding in 3 target cities.
- Onboard a certified cybersecurity consultant for credit ledger integrity review and DPDPA 2023-compliant data protection architecture audit.

RACI Matrix (Responsibility Assignment Matrix)

R = Responsible (performs the work) | A = Accountable (owns the outcome) | C = Consulted (provides input) | I = Informed (kept up to date) | A/R = Accountable and Responsible (acceptable in small flat teams)

Activity / Deliverable	CEO / Founder	Project Head	ML / Dev	Design Lead	QA Lead	Community Lead
Project Charter & Scope	A	R	C	C	I	I
Stakeholder Management	A	R	I	I	I	C
WBS Development	A	R	C	C	I	I
Schedule & Budget	A	R	C	C	I	I
Matchmaking Algorithm Design	C	C	A/R	I	I	I
UI/UX Design & Prototyping	C	I	I	A/R	I	I
Core Platform Development	C	R	A	C	I	I
Credit Ledger & Security	C	C	A/R	I	C	I
QA & Testing	C	R	C	I	A	I
Community Outreach & Training	A	C	I	I	I	R
Employer Partner Onboarding	A	R	I	I	I	C
Risk & Change Management	A	R	C	C	C	I
Go-Live Decision & Sign-Off	A	R	C	C	C	I

Staff Release Plan

- Development team members are released after successful production deployment and a 30-day post-launch stability monitoring period.
- Structured knowledge transfer, complete system documentation, and formal handover sessions completed before any staff transition.
- High-performing contributors with platform domain knowledge offered advisory or part-time roles in the Phase 2 national scale-up.

Training Plan

Training Area	Topics Covered	Target Audience	Timing
Technical Training	Skill Credit ledger operations, matchmaking algorithm monitoring, certificate system administration, CI/CD pipeline usage	Dev Team, DevOps	Week 1 (Onboarding)
Community Facilitation	Methodology for running effective peer skill exchange sessions, conflict resolution, learning outcome measurement	Community Trainers, NGO Partners	Week 3–4 (Pre-Pilot)
Safety & Ethics	DPDPA 2023 youth data privacy protocols, anti-exploitation safeguards, fair Skill Credit policy enforcement	All Team Members	Week 1 (Onboarding)
Soft Skills &	Slack, Notion, GitHub project	All Team Members	Week 1 (Onboarding)

Collaboration	management; community engagement best practices; stakeholder communication		
Security Awareness	OWASP Top 10, secure coding practices, incident response protocol, data breach notification procedure	Dev Team, QA Team	Week 2 (Sprint 1)

“Phase 1 focuses on a prototype-level implementation within academic constraints.”

Performance Review Framework

- Reviews conducted bi-weekly by the Head of Project with structured input from the Founder and relevant functional leads. Conducted at the end of each two-week sprint.
- Key Performance Indicators evaluated: matchmaking accuracy rate; credit system error rate per 1,000 transactions; user onboarding speed; certificate issuance rate; employer partner acquisition count per week.
- End-of-sprint retrospectives used to identify performance gaps, recognise outstanding contributions, and calibrate team resourcing for the following sprint.

Regulation and Policy Compliance

- Digital Data Protection: Full compliance with India's Digital Personal Data Protection Act (DPDPA) 2023; zero tolerance for user data sale or unauthorised third-party sharing.
- Youth Safety: Anti-exploitation safeguards built into Skill Credit terms; no user may extract unpaid labour under the guise of skill exchange; session caps enforced for new users.
- Occupational Licensing: Skills requiring formal government licensing (electrical installation, medical practice, legal services) are clearly flagged in the platform taxonomy and excluded from the peer certification scope.

21. Required Resources

Hardware, Software & Infrastructure

S.No	Category	Description / Details
1	Platform Infrastructure	AWS/GCP cloud hosting; PostgreSQL database cluster; Redis in-memory cache; API gateway; CI/CD pipeline (GitHub Actions); mobile app build environments
2	Matchmaking & AI Systems	TensorFlow Lite matchmaking model; NLP-based skill profiling tools; A/B testing framework; learning path recommendation engine; model retraining pipeline
3	Credit Ledger & Security	Immutable transaction ledger with AES-256 encryption; OWASP-compliant security toolkit; automated fraud detection module; audit trail system
4	UI/UX & Community Tools	Figma Professional (interface prototyping); React Native component library; WCAG 2.1 accessibility audit tools; offline-capable Progressive Web App (PWA); community training management app
5	Testing & Compliance Tools	Jest & Postman (unit/API testing); OWASP ZAP (penetration testing); k6 (load testing); DPDPA compliance checklist; educational content quality audit templates
6	Video Library Infrastructure	AWS S3 video storage; CloudFront CDN for low-latency streaming; video compression pipeline; offline download capability for low-bandwidth users

Human Resources Summary

Role	Quantity	Type	Duration	Key Skills
Full-Stack Developer	3	Contractual	3 months	React Native, Node.js, PostgreSQL, REST APIs
ML Engineer	1	Freelance	2 months	TensorFlow Lite, Python, NLP, recommendation systems
UI/UX Designer	1	Freelance	1.5 months	Figma, WCAG 2.1, React Native handoff, accessibility
System Architect	1	Contractual	1.5 months	Microservices, API design, DB schema, cloud architecture
Backend Developer / Security	2	Contractual	2.5 months	Node.js, Redis, AES-256, OWASP, fraud detection
QA Tester	1	Contractual	2 months	Jest, Postman, test case design, regression testing
Cybersecurity Specialist	1	Consultant	1.5 months	OWASP ZAP, penetration testing, DPDPA compliance
DevOps Engineer	1	Contractual	1 month	AWS, GitHub Actions CI/CD, k6, uptime monitoring
Community Facilitator	2	NGO Contract	1 month	Facilitation, community engagement, local language skills

Content Designer	1	Part-time	1 month	Instructional design, competency assessment build
Business Dev Lead	1	Part-time	2 months	Employer relations, SME partnerships, negotiation
Project Manager	1	Core Team	3 months	PMBOK, Agile/Scrum, stakeholder management, budgeting

22. Communication Management Plan

The Communication Management Plan defines the information needs of all project stakeholders, the methods and frequencies of communication, and the escalation protocols for issues and risks. It is developed per PMBOK 6th Edition Section 10 (Project Communications Management).

Message / Report	Description	Report To	Method	Frequency	Sender
Platform Development Updates	Sprint progress, feature completion status, bug resolution summary	CEO & Project Head	Daily Standup (15 min), Weekly Sprint Review	Weekly	Development Team
Matchmaking & Credit System Status	Algorithm performance metrics, credit transaction accuracy, system integrity indicators	Project Head & QA Lead	Technical Reports, Live Demo Sessions	Bi-Weekly	ML & Backend Team
Community Engagement Progress	Pilot user growth numbers, skill exchanges completed, offline meetup attendance outcomes	CEO & Community Lead	Campaign Dashboards, Field Briefings	Weekly	Community Coordinator
Quality & Testing Reports	Bug count by severity tier, test coverage %, matchmaking accuracy, security audit results	Project Head & QA Lead	QA Dashboards, Audit Logs	Bi-Weekly	Testing & QA Team
Employer Partner Updates	Partner onboarding status, job board activity, direct hiring outcome data	CEO & Business Dev Lead	Partner Reports, Email Updates	Monthly	Business Dev Team
Budget & Resource Usage	Expenditure vs. budget, resource utilisation rates, procurement milestone status	CEO & Finance Lead	Financial Reports, Budget Review Meetings	Monthly	Finance Team
Stakeholder Briefings	Project milestone summaries, social impact metrics, upcoming phase goals	All Stakeholders	Presentation Decks, Summary Reports	Monthly	Project Manager
Risk & Issue Escalation	New risks identified, triggered contingency plans, issues requiring stakeholder decision	CEO & Project Head	Risk Log, Escalation Email/Meeting	Within 24 hrs of identification	PM & QA Lead
Change	Proposed scope,	CEO (if >	Change	As each CR	Project

Request Notifications	budget, or schedule changes requiring review and approval	₹20k) / PM	Request Form (CR Template)	is submitted	Manager
NGO & Partner Field Updates	Community adoption progress, offline meetup outcomes, participant feedback from pilot cities	CEO & Community Lead	Field Visit Reports, Photo Journals	Bi-Weekly	Community Coordinator

23. Cost Estimation

Cost estimates are derived using bottom-up estimation per WBS Level-3 work package. Each work package is independently estimated before aggregation. The Contingency Reserve (₹50,000) covers identified risks. The Management Reserve (₹25,000) is held outside the cost baseline for unknown-unknown events. All amounts are in Indian Rupees (INR) inclusive of applicable GST.

1. Machinery and Infrastructure

S.No	Element	Estimated Cost (INR)
1	Cloud Server & Hosting Infrastructure (AWS/GCP) – 3 months	₹45,000
2	Android Device Testing Kit (5 QA test devices)	₹30,000
	Total Machinery Cost:	₹75,000

2. Raw Materials (Software and Tools)

S.No	Material	Estimated Cost (INR)
3	AI/ML Framework & API Licences (TensorFlow, OpenAI API tokens)	₹20,000
4	UI/UX Design & Prototyping Tools (Figma Professional – 3 months)	₹10,000
5	Video Streaming & Storage (AWS S3 + CloudFront – 3 months)	₹25,000
6	Security & Monitoring Tools (OWASP ZAP, Sentry, k6 load testing)	₹15,000
	Total Material Cost:	₹70,000

3. Manpower

S.No	Role	No.	Duration	Estimated Cost (INR)
7	Full-Stack Developer (×3)	3	3 months	₹1,20,000
8	ML Engineer (Matchmaking Algorithm)	1	2 months	₹1,00,000
9	UI/UX Designer (Freelance)	1	1.5 months	₹60,000
10	Community Facilitator / Trainer (×2)	2	1 month each	₹80,000
11	QA Tester & Security Analyst	1	2 months	₹70,000
	Total Manpower Cost:	₹4,30,000		

4. Core Development Modules

“All development module costs are included within manpower expenses. No separate allocation is made to avoid double counting.”

5. Cost Management Plan Summary

Estimating Methodology	Bottom-up estimation per WBS Level-3 work package. Each element independently estimated before aggregation (PMBOK 7.2).
Budget Baseline	₹5,75,000 (base cost) + ₹50,000 (Contingency Reserve) = ₹6,25,000 approved budget.
Contingency Reserve	₹50,000 (~8.7% of base cost). Released per risk event with PM approval and Risk ID documentation.
Management Reserve	₹25,000 for unknown-unknown events. Requires CEO approval and formal Change Request. Not included in project budget baseline.
Control Thresholds	Cost Variance (CV) and Schedule Variance (SV) alerts at $\pm 10\%$. Corrective action mandatory if $CPI < 0.90$.
Cost Change Authority	Changes $\leq ₹20,000$: PM approval. Changes $> ₹20,000$: CEO approval + formal Change Request.
EVM Reporting	Earned Value Analysis (EVA/EVM) performed at each milestone. CPI, SPI, EAC reported monthly to CEO.

6. Grand Total Estimated Budget

Base cost (Lines 1–3) = ₹5,75,000. Contingency Reserve (₹50,000) included. Total Project Budget = ₹6,25,000. Total project expenditure does not exceed ₹6,25,000

Line	Category	Total Cost (INR)
1	Machinery & Infrastructure	₹75,000
2	Raw Materials (Software & Tools)	₹70,000
3	Manpower (Contracted & Freelance)	₹4,30,000
4	Base Cost	₹5,75,000
5	Contingency Reserve (5.4% of base cost ₹5,75,000)	₹50,000
	GRAND TOTAL (Approved Cost Baseline)	₹6,25,000

24. Procurement Plan

Procurement Objectives

- Acquire all platform development tools, cloud infrastructure, and specialist talent within the ₹6,25,000 approved budget.
- Ensure timely procurement aligned with sprint schedules and quality benchmarks to avoid critical path delays.
- Select cost-effective and reliable vendors through transparent, competitive evaluation.
- Maintain full compliance with DPDPA 2023, open-source software licensing requirements, and NGO partnership agreements.

Procurement Methods

Method	Description	When Used
Direct Purchase	Purchase of standard, pre-defined tools and subscriptions without competitive bidding	Cloud credits, security libraries, mobile test devices, Figma licences
Request for Quotation (RFQ)	Soliciting competitive price quotes from multiple qualified vendors	Cloud hosting plans, video CDN, SMS gateway, monitoring tools
Request for Proposal (RFP)	Inviting structured technical and commercial proposals for complex deliverables	AI matchmaking engine, certificate generation module, employer job board
Freelance Milestone Contracts	Short-term engagement with phased, milestone-based payment schedules	ML Engineer, UI/UX Designer, Community Trainers
NGO Sub-contracts	Partnership agreements with established NGOs for community facilitation	Community Facilitators in 3 pilot cities

Equipment, Tools and Infrastructure

Item	Duration	Est. Cost	Method	Vendor Type
Cloud Hosting (AWS/GCP)	3 months	₹45,000	RFQ	Cloud Provider (AWS/GCP)
Video Streaming (S3 + CloudFront)	3 months	₹25,000	RFQ	AWS
Figma Professional	3 months	₹10,000	Direct Purchase	Figma Inc.
Security Tools (ZAP, Sentry, k6)	One-time	₹15,000	Direct Purchase	Open Source / SaaS vendors
Mobile QA Test Devices (×5)	One-time	₹30,000	Direct Purchase	Electronics Vendor

Selection Criteria: Competitive pricing; proven integration compatibility; data residency within India; responsive SLA support.

Manpower and Freelancers

Role	Duration	Est. Cost	Contract Type	Vendor Source
Full-Stack Developer (×3)	3 months	₹1,20,000	Milestone Contract	Developer Agency / Upwork
ML Engineer	2 months	₹1,00,000	Freelance Contract	AI/ML Specialist Platform
UI/UX Designer	1.5 months	₹60,000	Freelance Contract	Design Freelancer / Toptal
Community Facilitator (×2)	1 month each	₹80,000	NGO Sub-contract	NGO Partner Organisations
QA & Security Analyst	2 months	₹70,000	Milestone Contract	QA Specialist Firm

Selection Criteria: Relevant prior experience in social-impact technology; verifiable work samples; milestone delivery commitment; background verification clearance for youth-facing roles.

Vendor Evaluation Criteria

Criterion	Description	Weight
Technical Competency	Vendor must demonstrate relevant prior work (social impact tech, edtech, or equivalent platform)	40%
Cost Competitiveness	Total cost must be within ±15% of the budgeted line item for the category	25%
Delivery Reliability	Past on-time delivery record; milestone-based contract structure accepted	20%
Compliance & Security	Vendor must agree to DPDPA data handling requirements; no sub-contracting of sensitive modules without approval	10%
Support & Availability	Vendor must commit to response times ≤ 24 hours for critical issues during the project period	5%

25. Vendor Selection

Vendor selection follows a structured 8-step process designed to ensure objectivity, compliance, and value for money. All selection decisions are documented in the procurement log and approved by the PM or CEO depending on contract value.

Step	Activity	Description	Responsible	Timing
1	Market Research	Identify top vendors in edtech, social impact tech, and mobile platform development categories. Review portfolios, case studies, and client testimonials.	Project Head, Business Dev Lead	Week 1–2
2	Prequalification	Issue RFI (Request for Information) to shortlisted vendors. Assess compliance record, DPDPA data handling policies, and platform experience.	PM, Compliance Officer	Week 2–3
3	RFQ / RFP Issuance	Distribute structured RFQ/RFP documents to a minimum of 3 qualified vendors per category. Specify deliverables, timelines, and quality expectations.	Project Head	Week 2–3
4	Proposal Evaluation	Score proposals against the weighted evaluation criteria (Technical 40%, Cost 25%, Reliability 20%, Compliance 10%, Support 5%). Shortlist top 2 per category.	Evaluation Committee	Week 4
5	Negotiation & Due Diligence	Conduct reference checks with 2 prior clients; negotiate milestone payment terms, penalty clauses, and IP ownership. Obtain insurance certificates where applicable.	PM, Business Dev Lead	Week 4
6	Contract Award	Execute milestone-based or fixed-fee contracts with formal Statement of Work (SoW) attachments, DPDPA data processing agreements, and penalty clauses for delays.	CEO / PM	Week 5
7	Onboarding & Kickoff	Conduct vendor onboarding: grant repository access, introduce team protocols, confirm sprint schedules, and validate development environment setup.	Project Head, DevOps	Week 6
8	Performance Monitoring	Track vendor delivery via GitHub (code milestones), Trello (task completion), and weekly progress standup meetings. Issue Penalty Notice for SLA breaches.	Project Head	Ongoing

26. Procurement Timeline

The procurement timeline is aligned with the project schedule to ensure that all vendors are contracted and onboarded before their activities appear on the critical path. Any procurement delay beyond Week 5 would risk the March 24 development kickoff milestone.

Period	Calendar Dates	Phase	Responsible	Key Deliverable
Week 1–2	Feb 15 – Feb 28	Market Research & RFI	Project Head, Business Dev Lead	Vendor shortlist and qualification matrix prepared
Week 2–3	Feb 24 – Mar 7	RFQ / RFP Issuance & Collection	Project Head	Formal RFQ/RFP documents issued; quotes/proposals received from ≥ 3 vendors per category
Week 4	Mar 10 – Mar 14	Proposal Evaluation & Negotiation	PM, Evaluation Committee	Scoring matrix completed; top vendor per category identified; commercial terms negotiated
Week 5	Mar 17 – Mar 21	Contract Signing & Onboarding Prep	PM, CEO	All contracts signed; SoW attachments finalised; DPDPA data processing agreements executed
Week 6	Mar 24 – Mar 28	Development Kickoff	Project Head, DevOps	Vendor access granted; development environment validated; Sprint 1 formally commenced
Month 2 (Ongoing)	Apr 1 – Apr 18	Active Development Monitoring	Project Head	Bi-weekly vendor check-ins; milestone deliverable reviews; issue escalation as required
End of Month 3	Apr 2 – Apr 16	Final Payments & Close-Out	PM, Finance Lead	Milestone acceptance confirmed; final invoices approved; vendor performance documented

27. Project Change Control

All changes to the project's approved scope, schedule, budget, quality standards, or resource plan must follow the formal Change Control Process per PMBOK 6th Edition Section 4.6 (Perform Integrated Change Control). No change is implemented without documented sign-off from the appropriate authority.

Change Control Process

- Step 1 – Change Request Submission: Requestor completes the Change Request Template (CR Form) with all required fields.
- Step 2 – Impact Analysis: Project Manager conducts impact analysis across scope, schedule, budget, quality, and risk dimensions.
- Step 3 – Review & Approval: PM approves changes ≤ ₹20,000. CEO approval required for changes > ₹20,000. All decisions documented in the Change Log.
- Step 4 – Implementation: Approved changes are incorporated into the project plan, schedule, and budget baseline. WBS and risk register updated as needed.
- Step 5 – Communication: All affected stakeholders notified of the approved change, its impacts, and the revised baseline.

Change Request Template

CR Number	CR-[Auto-Assigned Sequential Number, e.g., CR-001]
Date of Request	[DD/MM/YYYY]
Requested By	[Name & Role]
Change Category	Scope Change Schedule Change Budget Change Quality Change Risk Response Resource Change
Change Description	[Detailed description of the proposed change, including which WBS element(s) are affected]
Business Justification	[Why is this change necessary? What problem does it solve or what opportunity does it address?]
Impact on Scope	[Additions, deletions, or modifications to the defined project scope]
Impact on Schedule	[Working days added or removed; effect on critical path activities and milestone dates]
Impact on Budget	[Cost increase or decrease in INR; funding source — Contingency Reserve, Management Reserve, or new approval]
Impact on Quality	[Effect on quality standards, KPIs, testing requirements, or acceptance criteria]
Impact on Risk	[New risks introduced or existing risks resolved / mitigated by this change]
Reviewed By	[Project Manager / Project Head — Date of Review]
Approved / Rejected By	[CEO if budget impact > ₹20,000 PM if budget impact ≤ ₹20,000]
Implementation Date	[DD/MM/YYYY]
Status	Pending Review Approved Rejected Deferred to Phase 2

Approved Change Log

Cumulative contingency consumed by all approved changes: CR-002 (₹15,000) + CR-003 (₹20,000) + CR-004 (₹15,000) = ₹50,000. The contingency reserve is now fully committed. Any further budget-impacting changes require access to the Management Reserve (₹25,000) with CEO approval.

CR #	Change Title	Budget Impact	Schedule Impact	Approved By	Approval Date	Status
CR-001	Matchmaking Algorithm Enhancement	₹0	+ 2 days to Activity 1.3	Project Head	Mar 15, 2026	Approved
CR-002	Credit Ledger Security Upgrade	₹15,000 (contingency)	+ 1 day	CEO	Mar 22, 2026	Approved
CR-003	UI/UX Accessibility Improvements	₹20,000 (contingency)	+ 3 days to Activity 1.2	CEO	Mar 28, 2026	Approved
CR-004	Employer Portal Scope Addition	₹15,000 (contingency)	0 (parallel sprint)	CEO	Apr 5, 2026	Approved

Change Narratives

CR-001 – Matchmaking Algorithm Enhancement

The initial matchmaking engine used keyword-matching only. Following internal testing, the algorithm was upgraded to a weighted multi-factor model incorporating skill proficiency level, geographic proximity, mutual availability, and historical endorsement ratings. Session completion rates improved by 22% in validation testing. Budget: ₹0 (within existing ML engineer scope). Schedule: +2 days to Activity 1.3 (absorbed within the 20-day window; no critical path impact).

CR-002 – Skill Credit Ledger Security Upgrade

Internal fraud audit identified vulnerabilities in the single-point session verification design. The ledger was redesigned with mutual verification (both parties confirm independently), automated anomaly detection, and cryptographic hash anchoring per transaction. Budget: ₹15,000 from contingency reserve. Schedule: +1 day.

CR-003 – UI/UX Accessibility Improvements

Pilot testing with underserved community users revealed a 39% task failure rate. Dashboard redesigned with icon-first navigation, Hindi/Telugu language support, 3-step session logging (reduced from 7), and offline caching. Task completion improved from 61% to 89%. Budget: ₹20,000 from contingency. Schedule: +3 days to Activity 1.2.

CR-004 – Employer Portal Scope Addition

Employer partners requested the ability to post targeted skill requirements to the SkillBarter community. A lightweight employer request board was added via formal CR-004. Executed in parallel sprint with no critical path delay. Budget: ₹15,000 from contingency. Schedule: 0 days impact.

28. Project Monitoring and Controlling

The Monitoring and Controlling Process Group ensures the SkillBarter project remains aligned with its approved scope, budget, quality standards, and social impact milestones throughout the entire project lifecycle. It operates continuously from project initiation through to closure, per PMBOK 6th Edition Section 4.5 (Monitor and Control Project Work) and Section 4.6 (Perform Integrated Change Control).

1. Performance Tracking and Schedule Control

- **Milestone Reviews:** Bi-weekly formal milestone assessments covering all seven project milestones. Each milestone requires a sign-off document before the next phase commences.
- **Sprint Reviews:** Sprint deliverables reviewed at the end of every 2-week sprint. A sprint is only closed when the Definition of Done for all committed stories has been met.
- **Progress Reports:** Sprint-based status reports submitted by development, QA, and community teams at the close of each sprint. Reports cover completed work, blockers, and next-sprint priorities.
- **Task Tracking Tools:** GitHub Projects (development tasks) and Trello (cross-functional tasks) provide real-time visibility into sprint backlog, work-in-progress, blockers, and completion status.
- **Gantt Chart:** Maintained and updated weekly to visualise phase dependencies, resource allocation, and schedule adherence against the approved baseline.
- **Earned Value Analysis (EVA/EVM):** CPI and SPI calculated at each milestone using Planned Value (PV), Earned Value (EV), and Actual Cost (AC). Corrective action is mandatory if CPI falls below 0.90.

2. Quality Assurance and Validation

- **Matchmaking Quality:** Algorithm output reviewed weekly against the 85% satisfaction target. Model retraining triggered automatically when accuracy drops below threshold.
- **Credit Ledger:** Daily automated reconciliation with weekly manual spot-checks by the security engineer. Any discrepancy > 0.01% triggers a security incident report.
- **Dashboard Validation:** Frontend performance tested across minimum 5 Android device profiles after every deployment for speed, accuracy, and offline functionality.
- **Community Feedback:** Structured feedback collected from pilot participants after every 10 completed skill exchanges. Aggregated into monthly quality audit reports.

3. Risk Monitoring and Mitigation

- **Risk Log:** Maintained and reviewed at every bi-weekly milestone meeting. All trigger events escalated to the PM within 24 hours of detection.
- **Residual Risk Review:** After each risk response is applied, the residual risk rating is re-evaluated to confirm it has fallen to the target level.
- **Risk Escalation Threshold:** Risks rated High × High are escalated to the CEO immediately. Risks rated Medium × High or High × Medium require PM action within 48 hours.

4. Cost and Resource Control

- **Weekly Expenditure Tracking:** All cloud costs, freelancer invoices, and tool subscriptions tracked weekly against the approved budget baseline.
- **Monthly EVM Review:** CPI, SPI, Estimate at Completion (EAC), and Variance at Completion (VAC) reported to the CEO each month.

- Contingency Status: Contingency reserve (₹50,000) fully committed after CR-001 through CR-004. All further unplanned spend requires Management Reserve access (₹25,000; CEO approval).

5. Communication and Stakeholder Engagement

- Daily Standups: 15-minute development and community team syncs for rapid issue identification and coordination.
- Weekly Sprint Review: Formal deliverable approval meeting chaired by the Project Head.
- Monthly Stakeholder Briefing: End-of-month presentation deck sent to all stakeholders including NGO partners, employer partners, and employment exchanges.

6. Change Management

- All proposed changes submitted via the Change Request Template before implementation. Decisions logged in the Change Log.
- Approval Authority: PM approves ≤ ₹20,000. CEO approves > ₹20,000. Emergency changes may be verbally authorised by the CEO with written confirmation within 24 hours.

7. Final Control Checkpoints

- Pre-Deployment Checks: Matchmaking accuracy ≥ 85%, credit ledger error rate < 0.01%, all security audit findings resolved, employer portal functional with 20 active partners.
- User Acceptance Testing (UAT): Formal UAT with 30 selected pilot participants (diverse demographics including low-literacy users). Minimum task completion rate of 85% required for go-live authorisation.
- Post-Launch Monitoring: Real-time analytics dashboard, daily community coordinator field reports, and weekly employer partner satisfaction check-ins maintained for 30 days post-launch.
- Lessons Learned: Formal lessons-learned workshop conducted within 2 weeks of project close. Outputs documented and archived as Phase 2 planning inputs.

WEBSITE

Why SkillBarter?



AI Matchmaking

Get matched with the perfect skill partner based on compatibility and interests.



Skill Credits

Teach to earn credits, spend credits to learn. A fair time-banking exchange system.



Peer Learning

Learn directly from skilled peers in personalized one-on-one sessions.



Job Opportunities

Access job listings tailored to your newly acquired skills.



Create an account

Start exchanging skills on SkillBarter

Full Name

Email

Password

Confirm Password

[Create Account](#)

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AJ

Alex Johnson

45 credits

Recent Transactions

Taught React.js to Rahul Verma

2024-04-18

+5

Learned UI/UX Design from Priya Sharma

2024-04-16

-5

Taught Node.js to Karan Singh

2024-04-15

+6

Learned Public Speaking from Sneha Patel

2024-04-12

-4

Taught React.js to Anita Desai

2024-04-10

+5

Upcoming Sessions

UI/UX Design Principles

Pending

Learning — with Priya Sharma

2024-04-20

10:00 AM

5 credits

Public Speaking Mastery

Accepted

Learning — with Sneha Patel

2024-04-22

2:00 PM

4 credits

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Full-time

Part-time

Internship

Freelance

Junior Frontend Developer

Full-time

TechCorp India

Build responsive web apps using React.js.

React.js

JavaScript

CSS

Mumbai, India

₹4-6 LPA

2024-04-10

UI/UX Design Intern

Internship

DesignHub

Assist in creating user-centered designs for mobile and web.

Figma

UI/UX Design

Remote

₹15K/month

2024-04-12

B. Tech Project Management Workbook | SkillBarter Social Tech Initiative | February–April 2026

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